Ex. No : 2 Depth First Search

AIM:

To implement depth first search codes) to traverse a graph and find a path from a start node to a goal node.

PROGRAM:

```
graph = {
    'A' : ['B', 'C'],
    'B' : ['D', 'E'],
    'C' : ['F'],
    'D' : [],
    'E' : ['F'],
    'F' : []
}

def recursive(g, node, visited = None):
    if visited is None:
        visited = []
    visited.append(node)
    for nb in g[node]:
        if nb not in visited:
            recursive(g, nb, visited)
    return visited
```

```
def path(g, start, goal):
    stack = [(start, [start])]
    visited = set()
    while stack:
        node, path = stack.pop()
        if node == goal:
            return path
        if node not in visited:
            visited.add(node)
            for nb in reversed(g[node]):
                stack.append((nb, path+[nb]))
    return None
print("From A:", recursive(graph, 'A'))
print("Path A->F:", path(graph, 'A', 'F'))
```

RESULT:

The Depth first search program was implemented in python and executed successfully.